

Chemistry Lab C - Chemistry Lab - Rickards Invitational Div. C SATELLITE - 11-01-2025
Student Answer Sheet (/rickards/ES/AnswerSheet?tid=000G80)

Instructions (shown before students start the test)

There will be a dry lab! No materials necessary.

Introduction (shown after students start the test)

Good luck!



Classify and balance each of the following reactions.

1. (1.00 pts) $\text{C}_3\text{H}_7\text{OH}_{(l)} + \text{O}_{2(g)} \rightarrow \text{H}_2\text{O} + \text{CO}_2$

- ☐ A) Combustion
- ☐ B) Decomposition
- ☐ C) Metathesis
- ☐ D) Single replacement reaction
- ☐ E) Synthesis

2. (1.00 pts) Fill in the first three missing coefficients (do not input the CO₂ coefficient).

3. (1.00 pts) $\text{Cu}^+ + \text{Zn} \rightarrow \text{Zn}^{2+} + \text{Cu}$

- ☐ A) Synthesis
- ☐ B) Decomposition
- ☐ C) Redox
- ☐ D) Single displacement

4. (1.00 pts) Fill in the first three coefficients.

5. (1.00 pts) $_ \text{Fe}^{2+} + _ \text{Fe}^{3+} + _ \text{OH}^- \rightarrow _ \text{Fe}_3\text{O}_4 + _ \text{H}_2\text{O}$

- ☐ A) Synthesis
- ☐ B) Double displacement reaction
- ☐ C) Combustion
- ☐ D) Decomposition
- ☐ E) Single displacement reaction

6. (1.00 pts) Fill in the reactant coefficients

7. (1.00 pts) Fill in the product coefficients

8. (1.00 pts) Ethylene + Hydrogen bromide $\rightarrow$ 1-bromoethane

- ☐ A) Addition
- ☐ B) Metathesis
- ☐ C) Redox
- ☐ D) Substitution
- ☐ E) Combustion

9. (1.00 pts) Enter the coefficients for each compound in the reaction

Thermodynamics!

10. (1.00 pts) For the following questions -- to ---, decide whether the given conditions would have a spontaneous reaction or not.

During an endothermic chemical reaction, a gas is consumed and a liquid produced.

- ☐ A) Yes.
- ☐ B) No.
- ☐ C) Can't decide with given information.

11. (1.00 pts) A solid substance dissolves in water, releasing heat as it does so.

- ☐ A) Yes.
- ☐ B) No.
- ☐ C) Can't decide with given information.

12. (1.00 pts) Two liquids mix, neither absorbing nor releasing heat.

- ☐ A) Yes.
- ☐ B) No.
- ☐ C) Can't decide with given information.

13. (1.00 pts) A gas condenses to a liquid, releasing heat.

- ☐ A) Yes.
- ☐ B) No.
- ☐ C) Can't decide with given information.

14. (1.00 pts) During an exothermic chemical reaction, four moles of gaseous reactants are turned into two moles of gaseous products.

- ☐ A) Yes.
- ☐ B) No.
- ☐ C) Can't decide with given information.

15. (4.00 pts) A substance is 25.07% Fe, 32.83% Cl, and 43.10% O. Determine the empirical formula of this substance. Show all work.

Atomic mass Fe = 55.85 Amu

Atomic mass Cl = 35.45 Amu

Atomic mass O = 16.00 Amu

16. (5.00 pts)

60 g of a carbohydrate with the empirical formula CH_2O is dissolved in 0.250 Kg of pure water. The freezing temperature of this new solution is 2.976 degrees celsius lower than the pure water solution. Determine the molecular formula of this molecule. Show all work.

$K_f(\text{water}) = 1.86 \text{ K} \cdot \text{kg/mol}$

Atomic weight C: 12 amu

Atomic weight H: 1.0 amu

Atomic weight O: 16 amu

17. (1.00 pts)

You want to determine the molarity of an unknown solution of HCl. To do so, you add magnesium in excess to 20 mL of acid and measure gas generation. At 293 K and 1 atm, the generated gas takes up 0.361 L.

What would an appropriate, cost-effective piece of lab equipment be to measure the volume of gas generated?

18. (1.00 pts) Calculate the molarity of the HCl solution. Assume that gas generated by the reaction of Mg with H₂O is negligible.

$$R = 0.0821 \text{ L}\cdot\text{Atm}/(\text{mol}\cdot\text{K})$$

Assume no pressure change from the generated gas.

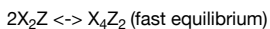
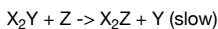
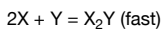
19. (1.00 pts) Explain why some substances are more soluble in cold water.

20. (1.00 pts)

Ce₂(SO₄)₃ is more soluble in cold water than hot water. Compared to dissolving Ce₂(SO₄)₃ in 10 degree C water, what happens when you dissolve it at 50 degrees C?

- ☐ A) It will dissolve faster, and more of it will dissolve
- ☐ B) It will dissolve faster, and less of it will dissolve
- ☐ C) It will dissolve slower, and more of it will dissolve
- ☐ D) It will dissolve slower, and less of it will dissolve

The steps of a reaction are hypothesized below.



All chemical species in the reaction are gaseous.

21. (1.00 pts) What is the rate law for the formation of X₄Z₂?

- ☐ A) rate = k[x][z][y]
- ☐ B) rate = k[x]²[z][y]
- ☐ C) rate = k[x]²[z]
- ☐ D) rate = k[x₂y]²
- ☐ E) rate = k[x₂y][z]

22. (2.00 pts) Show your work for the previous question.

23. (1.00 pts) In this reaction, Y would be a(n) _

- ☐ A) reactant
- ☐ B) product
- ☐ C) intermediate
- ☐ D) catalyst

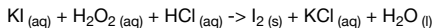
24. (1.00 pts) In this reaction, X_2Y would be a(n) _

- ☐ A) reactant
- ☐ B) product
- ☐ C) intermediate
- ☐ D) catalyst

25. (1.00 pts) Could a mixture of X and Z still form X_4Z_2 in the absence of Y?

- ☐ A) Yes
- ☐ B) No
- ☐ C) Maybe

The **unbalanced** reaction for the synthesis of iodine from potassium iodide is shown below.



26. (1.00 pts) What are the coefficients for the reactant side?

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27. (1.00 pts) What are the coefficients for the products?

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28. (3.00 pts) A student weighs out 67 grams of KI and dissolves it in 120 g 5% H_2O_2 (by mass) along with HCl in excess. Identify the limiting reactant. Show your work.

KI molar mass: 166.0 g

H_2O_2 molar mass: 34.0 g

29. (1.00 pts) Calculate the predicted yield of I_2 , in grams. Show your work.

I_2 molar mass: 253.8 g/mol

30. (1.00 pts) If the actual yield is 41 g, what is the percent yield?

31. (1.00 pts)

Another student performs the experiment and obtains a yield of 51 grams. What is the most likely contaminant? Note that the product was separated from the reaction via vacuum filtering.

- ☐ A) KI
- ☐ B) HCl
- ☐ C) H_2O_2
- ☐ D) KCl
- ☐ E) H_2O

32. (2.00 pts) The results of an experiment seeking to determine the rate law for a reaction between reactants A, B, and C are shown below.

Experiment	Reactant A conc. (M)	Reactant B conc. (M)	Reactant C conc. (M)	Rate (M/s)
A	2.0×10^{-2}	2.0×10^{-2}	2.0×10^{-2}	8.0×10^{-4} M/s
B	4.0×10^{-2}	2.0×10^{-2}	2.0×10^{-2}	8.0×10^{-4} M/s
C	8.0×10^{-2}	4.0×10^{-2}	2.0×10^{-2}	3.2×10^{-3} M/s
D	2.0×10^{-2}	2.0×10^{-2}	6×10^{-2}	2.4×10^{-3} M/s

Calculate the rate law for this equation.

- ☐ A) $\text{rate} = k[A][B][C]$
- ☐ B) $\text{rate} = k[B]$
- ☐ C) $\text{rate} = k[B][C]^2$
- ☐ D) $\text{rate} = k[A][B][C]^2$
- ☐ E) $\text{rate} = k[B]^2[C]$
- ☐ F) $\text{rate} = k[B][C]$

33. (2.00 pts) Calculate k, with units, for the previous question.

34. (2.00 pts) The results of an experiment determining the order of the decomposition of compound D are shown below. Determine the order of the reaction.

Time (s)	[D] (M)
0	0.500
30	0.389
60	0.303
120	0.184
300	0.0408
1200	2.23×10^{-5}

- ☐ A) Zero-order
- ☐ B) First order
- ☐ C) Second order

35. (2.00 pts) Calculate k for the reaction, rounded to three significant digits.

36. (3.00 pts) Which of the following statements about the speeds of gas molecules are true?

- I. At 100 K, the speed of a molecule of H_2 will equal the speed of a molecule of Cl_2 .
- II. The fastest gas molecule at 200 K will move slower than the average gas molecule at 500 K
- III. As temperature increases, the standard deviation of particle speeds increases.

- ☐ A) I only
- ☐ B) II only
- ☐ C) III only
- ☐ D) I and II only
- ☐ E) I and III only
- ☐ F) I, II, and III

37. (5.00 pts)

You have been tasked with assessing the concentration of lead in a body of fresh water. You have 100 mL of freshwater sample. Describe the process you would use to determine the concentration of lead. Assume it is the only heavy metal in solution. If you choose to use precipitation, assume lead left in solution is negligible.

You don't have to do any actual calculations! Just describe the process.

38. (1.00 pts)

Peroxydicarboxylic acids such as perbenzoic and peracetic acid decompose in solution by acid–base equilibria and bimolecular peracid–peroxide reactions. The observed rate law often depends strongly on pH. Kinetic analyses that assume the acid–base step is fast relative to subsequent bimolecular product-forming steps allow elimination of the intermediate using equilibria expressions. Consider a mechanism where the peracid $(HA) \rightleftharpoons A^- + H^+$ is fast, and $A^- + HA \rightarrow$ products (rate constant k_2) is rate-limiting. Which expression best represents an initial rate for product formation under the assumption $[HA]_0 \approx [HA] + [A^-]$ and the fast pre-equilibrium holds?

- ☐ A) $v_0 = k_2 * \frac{K_a}{[H^+] * [HA]_0^2}$
- ☐ B) $v_0 = k_2 * \frac{[HA]_0^2}{K_a [H^+]}$
- ☐ C) $v_0 = \frac{k_2 * [HA]_0^2}{[H^+]}$
- ☐ D) $v_0 = k_2 * \frac{K_a}{[H^+]} * [HA]_0^2$
- ☐ E) None of the above.

39. (1.00 pts)

Potential-energy surfaces, or PES, underlie all chemical reaction dynamics. Stationary points correspond to minima and saddle points correspond to transition states. Interpreting experimental rate data often requires understanding how a reaction path may bypass one saddle via an alternate valley–ridge inflection on the PES. Given a PES with two transition states (TS1 and TS2) in series connecting reactant $\rightarrow$ intermediate $\rightarrow$ product, but where dynamical bottlenecks cause most trajectories to proceed over TS2 even when TS1 is lower in energy. Which statement best reflects the conclusion when dynamics cause trajectories to favor a higher-energy TS2 over TS1 despite TS1 being lower in energy?

- ☐ A) The lowest-energy TS is always the rate-determining step, regardless of dynamics.
- ☐ B) Dynamical partitioning of trajectories on the PES — the higher-energy TS2 can control product formation even if TS1 is lower in energy.
- ☐ C) Because TS1 is lower, you can ignore dynamic effects; transition-state theory applies unchanged.
- ☐ D) Product branching is only determined by thermodynamics (product stabilities), not the PES saddle structure.
- ☐ E) None of the above.

40. (1.00 pts)

Classical transition-state theory (TST) can fail for reactions exhibiting trajectory bifurcation — where, after crossing a single transition state, reaction trajectories diverge dynamically toward multiple product wells without encountering distinct transition states. This “post-transition-state bifurcation” (PTSB) challenges the assumption that a unique dividing surface defines the reaction coordinate. For example, in pericyclic and rearrangement reactions such as the cyclopropylcarbinyl–homoallyl rearrangement or certain enzymatic Claisen rearrangements, quasi-classical trajectory simulations show that product ratios depend on vibrational excitation, mode-specific energy flow, and even isotopic substitution, despite similar barrier heights. In a reaction exhibiting a post-transition-state bifurcation on its potential-energy surface, which statement best explains why traditional transition-state theory fails to predict the correct product ratio?

- ☐ A) Because multiple reaction paths diverge after a single transition-state region, product selectivity also depends on dynamical momentum flow.
- ☐ B) Because the transition state lies above all product channels in energy, TST predicts no reaction at all.
- ☐ C) Because each product has a distinct pre-transition-state barrier, and TST neglects entropy differences.
- ☐ D) Because tunneling effects completely dominate product formation under all conditions.
- ☐ E) None of the above.

41. (1.00 pts)

Secondary kinetic isotope effects are most significant in hybridization and vibrational coupling. Heavy-atom KIEs (^{13}C , ^{15}N) have generally a KIE of <1.05 , but can be measured precisely with NMR and mass spectrometry. Computational methods also predict heavy-atom KIEs using vibrational partition function changes. Which conclusion is most consistent with a measured ^{13}C KIE of 1.014 at an sp^3 carbon undergoing substitution?

- ☐ A) The carbon center undergoes no change in bonding character.
- ☐ B) The reaction is entirely diffusion-controlled.
- ☐ C) Heavy-atom KIEs this small are always noisy and uninterpretable.
- ☐ D) The transition change is concerted but asynchronous in hybridization.
- ☐ E) None of the above.

42. (1.00 pts)

Although small in magnitude (typically 1.001–1.01), heavy atom isotope effects (e.g., $^{12}\text{C}/^{13}\text{C}$, $^{16}\text{O}/^{18}\text{O}$) provide subtle information about transition-state geometry and bonding rearrangements, like the secondary kinetic isotope effects. For instance, in enzymatic decarboxylation reactions, a measurable ^{13}C KIE can indicate partial bond cleavage at the carbon center in the rate-determining step. Why are heavy atom isotope effects generally much smaller than those involving hydrogen or deuterium?

- ☐ A) Heavy atoms have lower electron density and stronger nuclear shielding
- ☐ B) Zero-point energies between isotopes decreases sharply as atomic mass increases.
- ☐ C) Vibrational frequencies of heavy atoms dominate tunneling effects.
- ☐ D) Heavy isotopes undergo greater solvent stabilization.
- ☐ E) Because they do not affect any normal modes involved in bond cleavage.

43. (1.00 pts)

The Rice–Herzfeld mechanism describes radical chain oxidation of hydrocarbons: initiation, propagation, and termination. In a simplified chain mechanism, the steady-state concentration of radicals determines the overall rate. Let a hydrocarbon RH undergoing chain oxidation with initiation rate r_i (produces $\text{R}\cdot$) and propagation that produces product from $\text{R}\cdot + \text{O}_2$. If initiation is first-order in RH ($r_i \propto [\text{RH}]$) and termination is second-order in radicals ($\text{rate} = k_t [\text{R}\cdot]^2$), what is the scaling of the steady-state radical concentration $[\text{R}\cdot]$ with r_i and thus the overall oxidation rate (which is proportional to $[\text{R}\cdot]$)?

- ☐ A) $[\text{R}\cdot] \propto r_i$; $\text{rate} \propto [\text{RH}]$
- ☐ B) $[\text{R}\cdot] \propto (r_i/k_t)^{1/2}$; $\text{rate} \propto [\text{RH}]^{1/2}$
- ☐ C) $[\text{R}\cdot] \propto r_i^2$; $\text{rate} \propto [\text{RH}]^2$
- ☐ D) $[\text{R}\cdot]$ independent of r_i ; rate zero-order in $[\text{RH}]$
- ☐ E) None of the above.

44. (1.00 pts)

Michaelis–Menten kinetics models simple enzyme-catalyzed reactions: $\text{E} + \text{S} \rightleftharpoons \text{ES} \rightarrow \text{E} + \text{P}$. The rate is $v = \frac{k_{cat}[E]_0[S]}{K_M + [S]}$ where $K_M = \frac{(k_{-1} + k_{cat})}{k_1}$. Modern single-molecule and stochastic analyses show that deviations produce apparent higher-order kinetics or non-hyperbolic behaviors. If an enzyme has measured $K_M \approx 2 \mu\text{M}$ and $k_{cat} = 100 \text{ s}^{-1}$, but rapid-equilibrium binding experiments indicate K_d (true dissociation constant) = $0.02 \mu\text{M}$, what is the most consistent mechanistic inference?

- ☐ A) $k_{cat} \ll k_{-1}$, so $K_M \approx K_d$.
- ☐ B) $K_M = K_d$, $k_{cat} \gg k_{-1}$, so $K_M \approx k_{cat}/k_1$
- ☐ C) The enzyme is non-Michaelian and data invalid.
- ☐ D) The binding experiment must be wrong; discard it.
- ☐ E) None of the above.

45. (1.00 pts)

In the gas-phase isomerization of cyclopropane to propene, the reaction is observed to be first order with respect to cyclopropane. The Lindemann mechanism is proposed, where a high-energy intermediate is formed and decomposes to product. Which statement correctly relates molecularity to the observed kinetics?

- ☐ A) The reaction is bimolecular in the activation step and appears bimolecular under steady-state conditions.
- ☐ B) The reaction is unimolecular in both the mechanism and the observed kinetics.
- ☐ C) The rate-determining step is purely photochemical.
- ☐ D) The reaction is termolecular in all steps.
- ☐ E) None of the above.

46. (1.00 pts)

The oxidation of methanol on a gold surface proceeds via two parallel pathways:

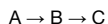
(1) $\text{CH}_3\text{OH} \rightarrow \text{HCHO}$ (k_1), and (2) $\text{CH}_3\text{OH} \rightarrow \text{CO}_2$ (k_2).

The ratio of products formed depends strongly on the surface temperature. If $k_1 = 2.0 \times 10^{-3} \text{ s}^{-1}$ and $k_2 = 8.0 \times 10^{-4} \text{ s}^{-1}$ at 350 K, what fraction of methanol converts to formaldehyde?

- ☐ A) 0.514
- ☐ B) 0.614
- ☐ C) 0.714
- ☐ D) 0.814
- ☐ E) There isn't enough information provided to solve this.

47. (1.00 pts)

Given a system where species A converts to B, which then converts to C:



Where, simultaneously, B can also revert to A (opposing reaction). Rate constants are k_1 ($\text{A} \rightarrow \text{B}$), k_{-1} ($\text{B} \rightarrow \text{A}$), and k_2 ($\text{B} \rightarrow \text{C}$). Which of the following correctly represents the steady-state concentration of the intermediate [B]?

- ☐ A) $[\text{B}] = (k_1[\text{A}]) / k_2$
- ☐ B) $[\text{B}] = (k_2[\text{A}]) / (k_1 + k_2)$
- ☐ C) $[\text{B}] = (k_1[\text{A}]) / (k_{-1} + k_2)$
- ☐ D) $[\text{B}] = (k_1[\text{A}]) / (k_1 + k_2 + k_{-1})$
- ☐ E) None of the above.

48. (1.00 pts)

In the photodissociation of formaldehyde, CTST predicted that H_2CO^* should cross a tight transition state, yielding $\text{H}_2 + \text{CO}$. However, trajectory simulations and spectroscopic data showed a surprising alternative: after C–H bond cleavage, the departing H atom “roams” far from the radical fragment, avoiding the conventional transition state, before abstracting another hydrogen atom to form H_2 . Which statement best describes the physical origin and kinetic consequence of a roaming mechanism compared to conventional transition-state pathways?

- ☐ A) The roaming mechanism involves quantum tunneling through a higher potential barrier, leading to faster reaction rates at low temperature.
- ☐ B) Roaming occurs on a shallow, anharmonic region of the PES where fragments avoid the conventional saddle point, leading to altered product energy distributions.
- ☐ C) Roaming trajectories correspond to vibrationally excited reactants recrossing the tight transition state multiple times before product formation.
- ☐ D) The roaming mechanism is characterized by direct head-on collisions forming a deep-bound intermediate with no change in rate law.
- ☐ E) None of the above.

49. (1.00 pts)

The Belousov–Zhabotinsky reaction is one of the most famous examples of a chemical oscillator, where concentrations of intermediates (such as Ce^{4+} and Ce^{3+} or bromomalonic acid species) periodically rise and fall. Instead of approaching a static equilibrium, the system cycles through oxidation and reduction states, producing periodic color changes. Mechanistically, the oscillations arise from coupled autocatalytic and inhibitory feedback loops. Which statement most accurately describes the kinetic and mechanistic features that allow oscillations to occur in such systems?

- ☐ A) Coupled nonlinear feedback of an autocatalytic step competing with an inhibitory step produces dynamic instability of the steady state, creating a limit cycle.
- ☐ B) The reaction exhibits oscillations only when all steps are first-order and thermodynamically reversible.
- ☐ C) Oscillations occur because the rate-determining step periodically reverses direction due to thermal fluctuations.
- ☐ D) The oscillation is caused by temperature-dependent diffusion of intermediates, not by chemical kinetics.
- ☐ E) None of the above.

50. (1.00 pts)

Innocent ligands, such as NH_3 or Cl^- , allow an unambiguous assignment of the metal oxidation state because they remain closed-shell and redox-inactive. In contrast, noninnocent ligands can participate in electron transfer with the metal center, resulting in ambiguous oxidation states and redox delocalization. For example, a nickel complex with a dithiolene ligand where redox activity was localized largely on the ligand, not the metal. Which of the following statements best explains the role of a noninnocent ligand in such systems?

- ☐ A) It acts as a purely σ -donor ligand that stabilizes high oxidation states of the metal without affecting redox behavior.
- ☐ B) It increases ligand field strength, thereby reducing metal–ligand covalency and preventing electron delocalization.
- ☐ C) It can accept electron density during redox cycling, causing the formal oxidation state of the metal to appear ambiguous while facilitating multi-electron reactivity.
- ☐ D) It remains inert during all catalytic steps but influences rate constants through steric hindrance.
- ☐ E) None of the above.

51. (1.00 pts)

Chelation refers to the binding of a single ligand to a metal center through multiple donor atoms, forming one or more five- or six-membered rings. This process dramatically increases complex stability in the so-called chelate effect. For example, the thermodynamic basis of EDTA binding to transition metals, noting that replacing monodentate ligands with multidentate ones results in a large increase in the K_f . This stability arises not merely from stronger bonds, but from entropic factors. Kinetic inertness can also increase, as chelation reduces the probability of simultaneous bond dissociation. Which statement best explains the thermodynamic origin of the chelate effect observed in metal–ligand complex formation?

- ☐ A) It arises primarily from stronger metal–donor σ -bonding in multidentate ligands compared to monodentate ligands.
- ☐ B) It results solely from π -backbonding interactions unique to aromatic ligands.
- ☐ C) It is due to the electrostatic attraction between metal cations and negatively charged ligands.
- ☐ D) It is mainly entropic, due to the replacement of several free ligands or solvent molecules by one multidentate ligand, increasing the number of free particles in solution.
- ☐ E) None of the above

52. (1.00 pts)

High-resolution microwave spectroscopy of interstellar molecules such as CO, HCN, and CH_3CN , reveal rotational transitions that encode precise information about molecular geometry. In diatomic molecules, the rotational energy levels are quantized as

$$E_J = B * J(J + 1),$$

where $B = h/(8\pi^2 I)$ is the rotational constant and I is the moment of inertia. Transitions follow the selection rule $\Delta J = \pm 1$ for rigid rotors, producing equally spaced lines in the spectrum. When isotopic substitution (e.g., $^{12}\text{C} \rightarrow ^{13}\text{C}$) occurs, the moment of inertia increases, affecting the transition spacing and thus allowing isotope identification in astrochemical spectra. If ^{12}CO shows a first rotational transition at 115 GHz, what qualitative change occurs when observing ^{13}CO ?

- ☐ A) The spacing decreases because the heavier isotope increases the moment of inertia, reducing the rotational constant B .
- ☐ B) The spacing between rotational lines increases because the molecule becomes lighter.
- ☐ C) The spacing remains unchanged since the rotational constant is independent of isotopic mass.
- ☐ D) The number of allowed rotational transitions doubles due to isotope coupling.
- ☐ E) None of the above.

- 53. (1.00 pts)** Infrared (IR) spectroscopy probes molecular vibrations where nuclei oscillate about equilibrium positions. For a diatomic molecule modeled as a harmonic oscillator
- $$E_v = (v + 0.5)h\nu,$$
- with selection rule $\Delta v = \pm 1$.
- In reality, ___ slightly modifies these levels and enables overtone transitions ($\Delta v = \pm 2, \pm 3 \dots$).
- A study on CO₂ and HCl demonstrated that isotopic substitution shifts vibrational frequencies due to changes in reduced mass μ , where
- $$\nu = \frac{1}{2\pi} * \frac{\sqrt{k}}{\mu}$$
- If HCl and DCl have nearly identical bond force constants (k), which of the following statements correctly describes their vibrational spectra?

- ☐ A) Both HCl and DCl have the same fundamental frequency since k is unchanged.
- ☐ B) DCl absorbs at a higher wavenumber because its reduced mass is smaller.
- ☐ C) DCl shows identical spacing between vibrational levels but shifted absolute intensities.
- ☐ D) DCl absorbs at a lower wavenumber because its greater reduced mass decreases vibrational frequency.
- ☐ E) None of the above.

- 54. (2.00 pts)** What terminology fills in the missing blank in the above question?

- 55. (1.00 pts)** A recent investigation on the decomposition of ozone (O₃) catalyzed by halogen radicals in the upper atmosphere is shown in the sequence:
- $$\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2 \text{ (fast)}$$
- $$\text{ClO} + \text{O} \rightarrow \text{Cl} + \text{O}_2 \text{ (slow)}$$
- Kinetic measurements showed that the rate of O₃ loss depended linearly on [O₃] but quadratically on [O].
- Which of the following differential rate laws correctly represents the overall rate of ozone decomposition under steady-state conditions?

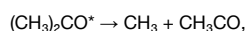
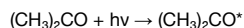
- ☐ A) rate = $k[\text{Cl}][\text{O}_3]$
- ☐ B) rate = $k[\text{O}_3][\text{O}]^2$
- ☐ C) rate = $k[\text{ClO}][\text{O}_3]$
- ☐ D) rate = $k[\text{O}_3]^2[\text{O}]$
- ☐ E) None of the above

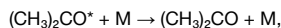
56. (2.00 pts)

The gas-phase recombination of NO and O to form NO₂. The experimental rate law was found to be rate = $k[\text{NO}][\text{O}][\text{M}]$, where M represents a third-body collision partner. What is the correct molecularity and interpretation of this elementary step?

- ☐ A) Unimolecular; spontaneous rearrangement
- ☐ B) Bimolecular; energy transfer from NO to O
- ☐ C) Termolecular; two reactive collisions plus an energy-stabilizing collision with M
- ☐ D) Bimolecular; catalyzed surface process
- ☐ E) None of the above.

- 57. (2.00 pts)** In the photochemical decomposition of acetone,





the concentration of the excited species $(\text{CH}_3)_2\text{CO}^*$ remains small.

Under steady-state approximation, the rate of radical formation is most sensitive to which parameter?

- ☐ A) Intensity of light only
- ☐ B) Pressure of inert gas M only
- ☐ C) Photon flux times excitation rate constant
- ☐ D) Sum of all rate constants
- ☐ E) None of the above

58. (2.00 pts)

In non-Markovian chemical kinetics of reactions occurring in solution, which of the following best describes the key difference from standard Markovian rate behavior?

- ☐ A) The rate constants depend only on temperature and not on time.
- ☐ B) The reaction dynamics exhibit memory effects due to solvent relaxation or coupling to slow environmental modes.
- ☐ C) The reactant concentration changes instantaneously in response to collisions, independent of solvent fluctuations.
- ☐ D) The reaction follows the Arrhenius law strictly, with no deviation over time.

59. (1.00 pts)

Ultrafast transient absorption spectroscopy studies reveal that geminate electron-hole recombination dynamics differ markedly between homogeneous molecular semiconductors and type-II heterojunctions such as polymer-fullerene blends. In homogeneous systems, geminate pairs typically recombine within picoseconds due to strong Coulombic attraction and limited charge separation length. In contrast, in heterojunctions, interfacial band offsets promote charge transfer, potentially suppressing geminate recombination.

Which of the following statements most accurately describes the fundamental kinetic distinction governing geminate recombination in homogeneous vs. heterojunction systems?

- ☐ A) In both systems, geminate recombination follows identical first-order kinetics, as recombination probability depends only on carrier density.
- ☐ B) In homogeneous systems, the geminate recombination rate is independent of the dielectric constant of the medium.
- ☐ C) Heterojunction systems accelerate geminate recombination because the interface facilitates rapid exciton diffusion into the bulk.
- ☐ D)

Homogeneous systems exhibit diffusion-controlled geminate recombination, whereas heterojunctions can exhibit kinetically hindered separation described by non-Markovian memory effects.

60. (1.00 pts)

Recent papers have investigated how interparticle interactions alter diffusion-controlled reaction dynamics, especially under confinement or near reactive boundaries. In such systems, the survival probability $S(t)$ of diffusing particles no longer decays exponentially, as assumed for independent random walkers, but often follows nontrivial power-law or stretched-exponential behaviors. These effects become prominent when diffusion is coupled with absorption or escape through partially reactive boundaries (finite reactivity k and when excluded-volume or electrostatic interactions between particles are significant.

Which of the following statements correctly captures the essential kinetic distinction introduced by interactions in survival and escape problems for diffusing particles?

- ☐ A) The mean survival time remains invariant under inclusion of interparticle interactions, as diffusion remains Markovian.
- ☐ B) Interparticle interactions lead to non-exponential survival probability decay and spatial correlations that modify the effective boundary reactivity and escape flux.
- ☐ C) When particles interact, the diffusion equation remains linear, so the escape probability is simply additive over all particles.
- ☐ D) Interactions increase the escape rate linearly with particle number due to higher collision frequency.
- ☐ E) None of the above.

61. (1.00 pts)

In classical Smoluchowski theory, diffusion-controlled reactions assume perfectly absorbing boundaries—reactants react instantaneously upon encounter. However, many modern experimental systems, such as enzyme–substrate encounters in crowded cytoplasmic environments or surface-catalyzed reactions in nanopores, deviate from this ideal behavior. For example, imperfect diffusion control arises when the intrinsic reactivity at the encounter surface is finite, leading to radiation boundary conditions where both diffusion and surface reaction kinetics determine the overall rate.

Which of the following statements correctly describes the kinetic consequence of imperfect diffusion control in such systems?

- ☐ A) The reaction rate constant interpolates between diffusion-controlled and activation-controlled limits depending on the ratio of intrinsic surface rate to the diffusive encounter rate.
- ☐ B) The system exhibits a purely exponential decay in reactant concentration, unaffected by boundary conditions.
- ☐ C) The overall rate constant is determined solely by diffusion and is independent of the surface reactivity.
- ☐ D) The mean encounter time between reactants decreases as the intrinsic surface reactivity decreases.
- ☐ E) None of the above.

We hope you enjoyed the exam!